

Engineering Report

To, AeroGrid CTO,

Here is a summary of the turbine results and conclusions deduced from the data. This report aims to show what the analysis script entail and how to re-architect the system

Findings

The specific turbines that fail according to the script are turbines 4 and 7. This is due to the fact that turbine 4 has an average temperature above 85 °C and turbine 7 has vibration level spikes above 15 mm/s. Both these are anomalous results and therefore they are ruled out of data processing and highlighted and failed turbines.

Architecture Justification

I chose a message-queue layer system to buffer the turbine telemetry as prevents crashes by separating the data producers from consumers, hence ensuring that sudden spikes don't overload the processing service. A serverless compute layer processes the data in parallel and eliminates the problems that caused the previous system to fail. Processed data is stored in a cloud object storage which is resilient, available and perfect for large datasets. This combination means there is a good fault tolerance and is reliable.

Cost Optimisation

One method to optimise cloud computing or storage costs is to implement tiered storage. This means that recent turbine data remains in hot storage for fast access whereas older turbine data can be move to archive tiers as the only purpose is historical analysis. As a result, the storge costs over a long period of time are reduced while effective performance is maintained. This a sustainable and easy fix to the massive, continuous data pipeline

I hope these solutions are applied to the AeroGrid as this improved system will solve the data failure in the system and ensure the turbine readings are processed and stored for a long time.

Sincerely,

Daniel Adebayo